



CS & IT ENGINEERING



Algorithms

Divide & Conquer

Lecture No.- 06



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Quick Sort

↳ Partition Algo

Topics to be Covered



Topic

Topic

Topic

Questions on Quick Sort
Matrix Multiplication.



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

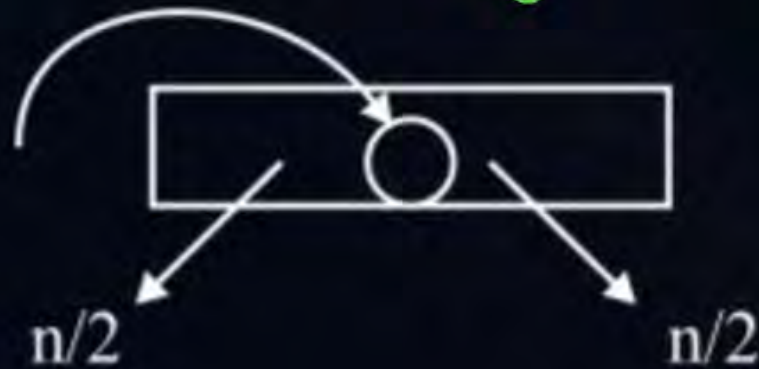


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Quick Sort :

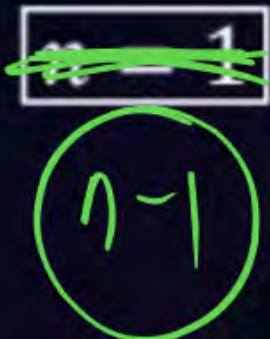
1. Time Complexity :

(1) Best case : Avg Case $O(n \log n)$: Unsorted input → (random)



$$T(n) = 2T(n/2) + O(n)$$

(2) Worst case : $O(n^2)$: Sorted input → (asc/disc)



$$T(n) = T(n-1) + O(n)$$

Apti

2	7	5	9	3
---	---	---	---	---

pivot :- 2, 9, 3, 7, 5 (Zigzag)

2	3	5	7	9
---	---	---	---	---

2	3	5	7	9
---	---	---	---	---



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Space Complexity :

1. Best Case : $O(\log n)$ → Recursion stack
2. Worst Case : $O(n)$ → ~~Spaced~~ Recursion stack
Skewed





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#Q. Let P be a quick sort program to sort numbers in ascending order. Let t_1 and t_2 be the time taken by the program for the inputs [1 2 3 4] and [5 4 3 2 1] respectively. Which of the following holds?

42%

- A** $t_1 = t_2$
- B** $t_1 > t_2$
- C** $t_1 < t_2$
- D** $t_1 = t_2 = 5 \log 5$

$t_1 \rightarrow i/p1 \rightarrow \underline{asc} : \underline{O(n^2)}, n=4$
 $t_2 \rightarrow i/p2 \rightarrow \underline{desc} : \underline{O(n^2)}, n=5$
 $t_1 < t_2$



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#Q. Let P be a Quick Sort Program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $\{1, 2, 3, 4, 5\}$ and $\{4, 1, 5, 3, 2\}$ respectively. Which one of the following holds.

- A** $t_1 = 5$
- B** $t_1 < t_2$
- C** $t_1 > t_2$
- D** $t_1 = t_2$

$t_1 \rightarrow i/p_1 \rightarrow \underline{\text{asc}} \rightarrow O(n^2)$ 56%

$t_2 \rightarrow i/p_2 \rightarrow \text{random} \rightarrow O(n \log n)$ $n=5$

(Avg)

$t_1 > t_2$



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#Q. Quick Sort is run on two inputs shown below to sort in ascending order taking first element as pivot

i. $1, 2, 3, \dots, n$

ii. $n, n-1, n-2, \dots, 2, 1$

Let C_1 and C_2 be the number of comparisons made for the inputs (i) and (ii) respectively. Then,

A

$$C_1 > C_2$$

B

$$C_1 < C_2$$

C

$$C_1 = C_2$$

D

We can't say anything for arbitrary n

$$\begin{aligned} C_1 &= i/p_1: \text{asc}(1 \rightarrow n) : \text{wc} \Rightarrow O(n^2) \rightarrow n = n \\ C_2 &= i/p_2: \text{desc}(n \rightarrow 1) : \text{wc} \Rightarrow O(n^2) \rightarrow n = n \end{aligned}$$



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V. imp $\rightarrow$ unique

#Q. Assume that merge sort takes 30 sec to sort 64 elements in worst case. What is the approximate number of elements that can be sorted in the Worst case using merge sort using 6 minutes?

44%

Ans: 512

$t = 30 \text{ sec}$ $n = 64 \text{ elem}$

WC

6 min

?

Merge Sort wc: $O(n \log n)$

Given
 $n = 64$
 $t = 30 \text{ sec}$

$$t = \underline{C} * n \log_2 n$$

$$30 = C * 64 * \log_2(64)$$

$$30 = C * 64 * \log_2(2^6)$$

$$30 = C * 64 * 6$$

$$C = \frac{30}{64 \times 6}$$

$$\Rightarrow \boxed{C = \frac{5}{64}}$$

$$t = 6 \text{ min}$$

$$= \underline{6 \times 60 \text{ sec}}$$

$$t = c \times n \log n$$

$$6 \times 60 = \frac{5}{64} \times n \log_2 n$$

$$\begin{array}{r} 5120 \\ - 512 \\ \hline 4608 \\ \hline \hline \end{array}$$

$$n \log_2 n = \frac{6 \times 60 \times 64}{5}$$

$$\boxed{n \log_2 n = 6 \times 12 \times 64}$$

$$n = 512$$

$$= 2^9$$

$$n \log_2 n = \underline{4608}$$

$$9 \times 512$$

$$\boxed{n = 512}$$



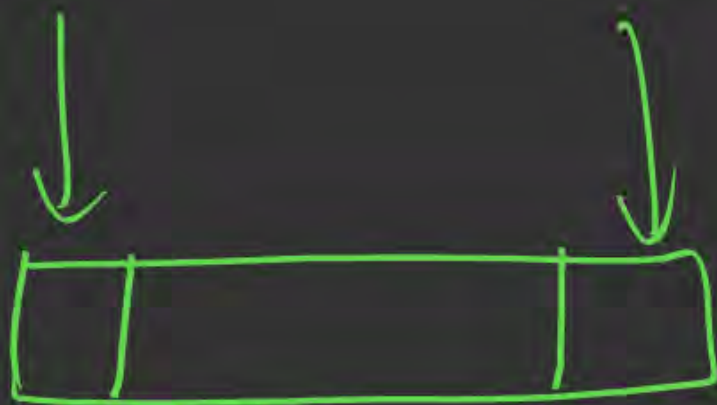
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#Q. An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.

$Val \times 100$

pe

pivot = min or max



$$P_{\text{rob}} = \frac{2}{25} = \boxed{0.08}$$



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Matrix Multiplication Problem

w.r.t. square matrix of size $n \times n$.



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Basics :

$$A_{n \times n} * B_{n \times n} \rightarrow C_{n \times n}$$

In general :

$$A_{r_1 \times c_1} * B_{r_2 \times c_2} = \underline{\underline{C_{r_1 \times c_2}}}$$


A and B are multiplicable only if $\boxed{c_1 = r_2}$ ✓



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Eq : Matrix Addition

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}_{2 \times 2}$$

$$A_{n \times n} + B_{n \times n} \rightarrow C_{n \times n}$$



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Time Complexity of Previous Method of Matrix Addition

$A_{n \times n} + B_{n \times n} \Rightarrow$ $\left\{ \begin{array}{l} \text{for } i : 1 \rightarrow n \\ \quad \text{for } j : 1 \rightarrow n \\ \quad \quad C[i, j] = A[i, j] + B[i, j] \end{array} \right.$

TC $\Rightarrow$ $O(n^2)$ [Matrix Addition]
and
Matrix subtraction



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Matrix Multiplication :

Given two square matrices each $\rightarrow n \times n$

$$\begin{matrix} & A & & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & * & \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} & = & \begin{bmatrix} \underline{1 \times 5 + 2 \times 7} & \underline{1 \times 6 + 2 \times 8} \\ \underline{3 \times 5 + 4 \times 7} & \underline{3 \times 6 + 4 \times 8} \end{bmatrix}_{2 \times 2} \end{matrix}$$



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r. imp

Matrix multiplication in general : (Non D and C)

$$\begin{matrix} & A & & B & & C \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} & * & \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} & = & \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \end{matrix}$$

Then, $t_1 \rightarrow$

{	$c_{11} = a_{11} * b_{11} + a_{12} * b_{21}$	$\rightarrow$	<u>1</u>	<u>2</u>	}	4 Addition & 8 Multi.
	$c_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$	$\rightarrow$	1,	2		
	$c_{21} = a_{21} \times b_{11} + a_{22} \times b_{21}$	$\rightarrow$	1,	2		
	$c_{22} = a_{21} \times b_{12} + a_{22} \times b_{22}$	$\rightarrow$	1,	2		



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Code of prev matrix multiplication logic (Non DnC)

$$A_{n \times n} * B_{n \times n} = C_{n \times n}$$

For $i : 1 \rightarrow n :$

$\Rightarrow$ $O(1)$ space

For $j : 1 \rightarrow n :$

For $k : 1 \rightarrow n$

$$C[i, j] \stackrel{+=}{=} A[i, k] * B[k, j]$$

Can ~~are~~ do
better using
DnC
approach >

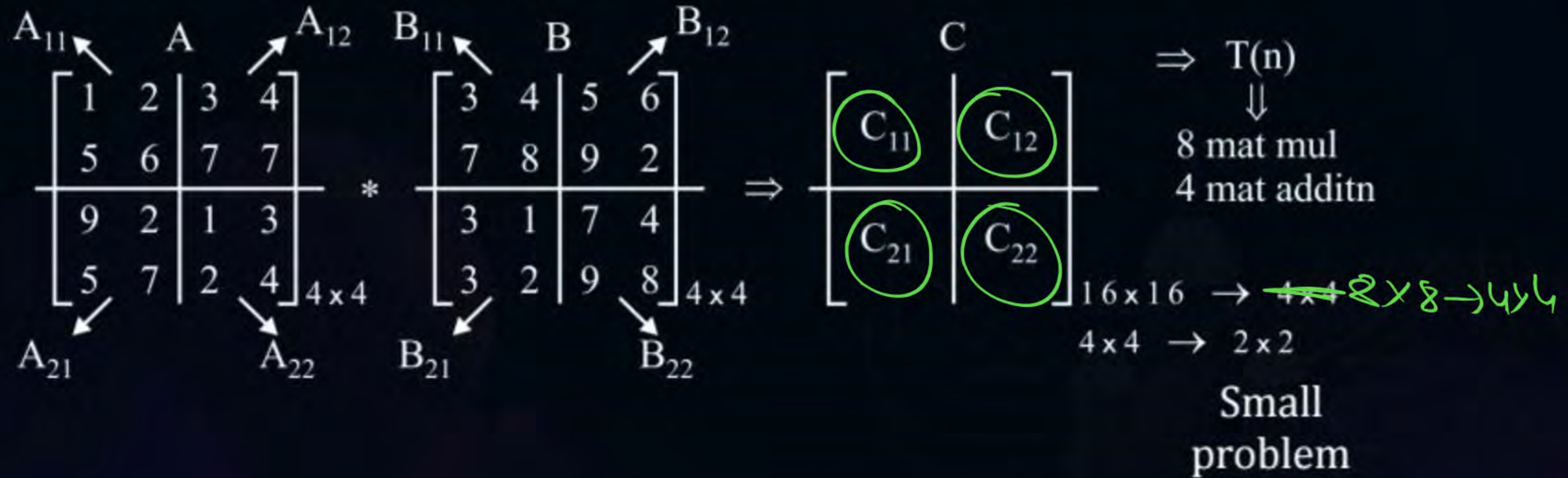
$\Rightarrow$

$$TC = O(n^3)$$



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Eg : Divide & Conquer based approach





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Sub-problem

Where,

$$C_{11} = A_{11} * B_{11} + A_{12} * B_{21} \quad (1)$$

$$C_{12} = A_{11} * B_{12} + A_{12} * B_{22} \quad (2)$$

$$C_{21} = A_{21} * B_{11} + A_{22} * B_{21} \quad (3)$$

$$C_{22} = A_{21} * B_{12} + A_{22} * B_{22} \quad (4)$$

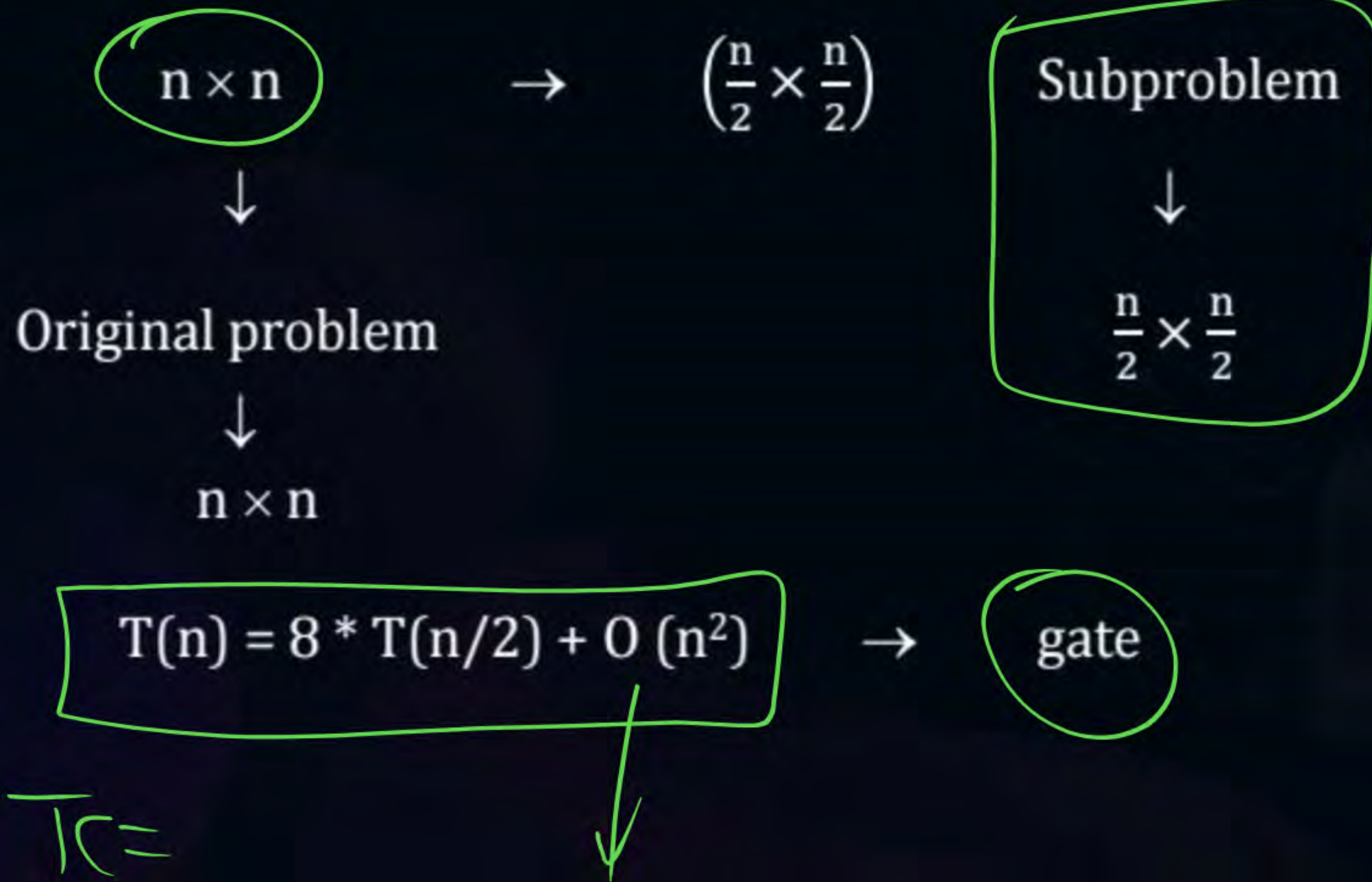
$A, B \rightarrow n \times n$

Add	Mul	
1,	2	4 sub-matrix addition & 8 sub-matrix Mul Subproblem → Divide
1,	2	
1,	2	
1,	2	



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Divide :





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Let $T(n)$ represent the time complexity to multiply two square matrices A & B each of size $n \times n$. (square matrix) $\rightarrow$ Recurrence for DnC based matrix multiplication.

$$\begin{aligned} * \quad T(n) &= C, \quad n \leq 2 \\ T(n) &= 8 T(n/2) + b * n^2, \quad n > 2, \quad b > 0 \end{aligned}$$

$$T(n/2) = 8T(n/2^2) + b * n^2/4$$

$$\begin{aligned} T(n) &= 8 \left[8T\left(\frac{n}{2^2}\right) + \frac{bn^2}{4} \right] + b * n^2 \\ &= 8^2 T\left(\frac{n}{2^2}\right) + 2bn^2 + bn^2 \\ &= 8^2 T\left(\frac{n}{2^2}\right) + 3bn^2 \quad \dots (2) \end{aligned}$$

$$2^k = n$$

$$(2^k)^3 = n^3$$

$$2^{3k} = n^3$$

$$(2^3)^k = n^3$$



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General Term :

$$8^k = n^3$$

$$T(n) = 8^k T\left(\frac{n}{2^k}\right) + (2^k - 1) * bn^2$$

... (2)

Let $\frac{n}{2^k} = 1 \Rightarrow 2^k = n \Rightarrow k = \log_2 n$

$2^k \rightarrow n$ $T(n) = n^3 T(1) + (n - 1) * bn^2$

$8^k \rightarrow n^3$ $= n^3 * c + (n - 1) * bn^2$

$= n^3 * c + bn^3 - bn^2 \Rightarrow \boxed{O(n^3)}$

TC after using this simple DnC based approach is also $O(n^3)$.



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Observations :

1. In DnC based matrix multiplication, there are 8 subproblems (sub-matrix multiplication), involved in getting C_{11} , C_{12} , C_{21} , C_{22} .
2. Time complexity will only be reduced, if the no. of sub-matrix multiplications are reduced from 8 to a smaller value.



Logic :

"STRASSEN" → Research

$$\left\{ \begin{array}{l} a * b = a + a + \dots + a \\ \qquad \qquad \qquad \underbrace{\hspace{2cm}} \\ \qquad \qquad \qquad \text{b times} \end{array} \right.$$



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Strassen's Matrix Multiplication

submatrices

$$\left\{ \begin{array}{l} P = (A_{11} + A_{22})(B_{11} + B_{22}) \\ Q = (A_{21} + A_{22})^* B_{11} \\ R = A_{11}^* (B_{12} - B_{22}) \\ S = A_{22}^* (B_{21} - B_{11}) \\ T = (A_{11} + A_{12})^* B_{22} \\ U = (A_{21} - A_{11})^* (B_{11} + B_{12}) \\ V = (A_{12} - A_{22})^* (B_{21} + B_{22}) \end{array} \right.$$
$$\left\{ \begin{array}{l} C_{11} = P + S - T + V \\ C_{12} = R + T \\ C_{21} = Q + S \\ C_{22} = P + R - Q + U \end{array} \right.$$

Result of Strassen's research on matrix multiplications.

$\{A_{n \times n} * B_{n \times n} \rightarrow C_{n \times n}\} \rightarrow$ Original problem

$$A_{ij}, B_{ij}, C_{ij} \Rightarrow \frac{n}{2} \times \frac{n}{2}$$

Additionally created/used sub-matrices

$$P, Q, R, S, T, V, U \Rightarrow \left(\frac{n}{2} \times \frac{n}{2} \right)$$



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Let $T(n)$ represents the TC for the matrix multiplication using Strassen's approach :

$$T(n) = c, \quad n \leq 2$$

$$T(n) = 7 \times T\left(\frac{n}{2}\right) + bn^2, \quad n > 2$$

Strassen's
Recurrence

$$T(n/2) = 7 * T\left(\frac{n}{2^2}\right) + b\left(\frac{n}{2}\right)^2$$

$$T(n) = 7 \left[7T\left(\frac{n}{2^2}\right) + \frac{bn^2}{2} \right] + bn^2$$

$$= 7^2 T\left(\frac{n}{2^2}\right) + \frac{7bn^2}{4} + bn^2$$

HW



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General Term :

$$T(n) = 7^k T\left(\frac{n}{2^k}\right) + bn^2 * \left(\sum_{i=0}^{k-1} \left(\frac{7}{4}\right)^i \right)$$

$$\text{GP : } \sum_{i=1}^n x^i < x^{n+1}$$

$$T(n) = 7^k T\left(\frac{n}{2^k}\right) + bn^2 * \left(\frac{7}{4}\right)^k$$

$$= 7^k \underline{T(1)} + bn^2 * \left(\frac{7}{4}\right)^k$$

$$= 7^k * c + \cancel{bn^2} * \cancel{\frac{7^k}{4^k}}$$

$$2^k = n$$

$$4^k = n^2$$

$$n/2^k = 1$$

$$2^k = n$$

$$k = \log_2 n$$

$$\begin{aligned} \sum_{i=1}^3 &= 2^1 + 2^2 + 2^3 \\ &= 2 + 4 + 8 \\ &= 14 < 2^k \\ &= 14 < 16 \end{aligned}$$



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$$T(n) = 7^k * c + bn^2 * \frac{7^k}{n^2}$$

$$T(n) = 7^k * c + 7^{12} * b$$

$$= 7^k (b + c)$$

$$T(n) = d * 7^k$$

$$T(n) = O(7^k)$$

$$T(n) = O(n^{2.81})$$

$$n^3 \Rightarrow n^{2.81}$$

Let

$$b + c = d$$

(constant)

$$k = \log_2 n$$

$$7^k = 7^{(\log_2 n)}$$

$$= n^{\log_2 7}$$

$$= n^{2.81}$$

$$\log_2 7 = 2.81$$

Handwritten diagram illustrating the relationship between k , n , and 7 :

$$k = \log_2 n$$

Diagram showing a tree structure with 7 at the root, branching into 7 children, and further branching into n children, illustrating the recursive process.



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Complexity Analysis of matrix mul (square) : Time complexity summary

- | | | |
|--------------------------------|---------------|-----------------------------------|
| 1. Non - DnC | $\Rightarrow$ | <u><u>$O(n^3)$</u></u> |
| 2. <u>Simple DnC</u> | $\Rightarrow$ | <u><u>$O(n^3)$</u></u> |
| 3. <u>Strassen's based DnC</u> | $\Rightarrow$ | <u>$O(n^{2.81})$</u> |

V. imp



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Space Complexity :

1. ~~Non~~ - DnC $\Rightarrow$ $O(1)$ $\rightarrow$ Constant
2. Simple DnC $\Rightarrow$ Recursion stack $\rightarrow$ $O(\log_2 n)$
3. Strassen's DnC $\Rightarrow$ $O(\log n + n^2) = O(n^2)$ $\rightarrow$ P, Q, R, S, R, U, V
 ~~$O(n^3) = O(n^{2.81})$~~



2 mins Summary



Topic

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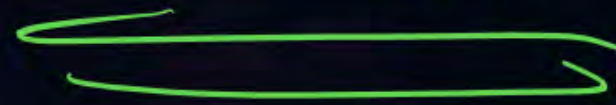
Topic

Topic

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Quick Sort

Matrix Mul





THANK - YOU

